Phase-Based Video Motion Magnification

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Abstract

This project implements a computational video processing technique to reveal and magnify subtle, imperceptible motions in standard video sequences. The core of this work is the phase-based approach to motion magnification, which leverages the properties of complex steerable pyramids. Input video frames are transformed into a multi-scale, multi-orientation representation, allowing for the isolation of local phase information.

Temporal variations in this phase data, which directly correspond to motion, are then temporally bandpass filtered to isolate specific frequencies of interest (e.g., a pulse or a vibration). The phase variations within this band are then amplified by a specified factor. To preserve video quality and prevent artifacts, the processing is applied selectively, often to the luminance (Y) channel of the YIQ colorspace, and may incorporate amplitude-weighted blurring and adaptive magnification to manage noise and large motions.

The final video is reconstructed from the modified phase information, resulting in a sequence in which the targeted subtle motions are exaggerated and made clearly visible.

1. Introduction

It is seen that many natural phenomena are too subtle to be perceived naturally by the human eye. Examples include pulsation of blood under the skin, rise and fall of an infant's chest during respiration as well as minute structural vibrations of a bridge under stress.However these motions are not absent but merely they are hidden within the pixel intensities of their video recordings.We can extract these motions using computational video motion magnification to amplify these subtleties algorithmically.

Earlier techniques like Eulerian Video Magnification (EVM) were proposed by Wu et al. This technique was based directly on the pixel intensity variation. it was effective for small motions but was sensitive to intensity changes and noise , leading to virtual artifacts. the phase based method introduced by Neal Wadhwa et al . leverages the relationship between local phase variations and sub pixel motion. Operation in the phase domain brings us greater accuracy and robustness , particularly for textured or complex scenes.

Our project implements the Phase-Based Video Motion Magnification algorithm to extract and amplify subtle motion signals embedded in video.In order to achieve this we created a pipeline of multiple stages.Firstly, the input video is converted to an appropriate color space(in our case YIQ) to separate luminance and chrominance minimizing color loss during magnification.Next, each frame is decomposed into a complex steerable pyramid, which represents spatial information across different scales and orientations.Local phase variations within this pyramid are tracked over time, and a temporal bandpass filter isolates frequencies corresponding to the motion of interest (e.g., 0.5–3 Hz for a human pulse). These phase changes are then selectively amplified before reconstructing the frames from the modified pyramid representation. The result is a video in which previously imperceptible motions become visually apparent.

This paper describes the underlying theory, implementation architecture, and the role of image processing in realizing this system, highlighting both the advantages and limitations of the phase-based approach.

2. Background

The field of computational video analysis seeks to extract latent information encoded in image sequences that may not be perceptible to the human visual system. Subtle motions, such as those caused by physiological processes, sound-induced vibrations, or material strain, are often present at amplitudes smaller than a single pixel. Conventional motion estimation techniques, including optical flow and block matching, explicitly estimate pixel displacements between consecutive frames. However, these methods are limited by noise sensitivity, quantization, and their inability to reliably detect sub-pixel motions. Our technique instead aims to reveal and amplify these imperceptible temporal variations directly within the video signal.

2.1. Eulerian Video Magnification

In this domain one of the first proposed techniques was Eulerian Video Magnification, credited to Wu et al. EVM adopts an *Eulerian* framework and proceeds by analyzing temporal intensity variation of each pixel while keeping its spatial coordinates constant.This is contrasted by a *Lagrangian* approach, which tracks each and every pixel's motion across frames.

Its working can be formalised in the following manner:

Let $I(x, y, t)$ denote the image intensity at pixel location (x, y) and time t. If the pixel undergoes a small motion $\delta(x, y, t)$, the intensity of the observed signal can be written as a displaced version of a reference frame $f(x, y)$:

$$I(x, y, t) = f(x + \delta(x, y, t), y). \tag{1}$$

For small displacements, a first-order Taylor expansion yields:

$$I(x, y, t) \approx f(x, y) + \delta(x, y, t)\frac{\partial f(x, y)}{\partial x}. \tag{2}$$

If $\delta(x, y, t)$ varies over time (for example, periodic motion due to a pulse), then each pixel's temporal intensity signal encodes information about that motion. EVM applies a temporal bandpass filter $B(\cdot)$ to isolate the desired frequency range:

$$\hat{I}(x, y, t) = B[I(x, y, t)], \tag{3}$$

and subsequently amplifies the filtered component by a scalar α:

$$\tilde{I}(x, y, t) = f(x, y) + (1+\alpha)\left(\hat{I}(x, y, t) - f(x, y)\right). \tag{4}$$

This model effectively amplifies periodic variations but suffers from several limitations.As amplification is applied directly to intensity, EVM is highly sensitive to lighting changes, color noise, and large motions that violate the small-motion assumption in Eq. 2. These artifacts manifest as unnatural color shifts and halo effects near edges and produce inaccuracies in the resulting video.

2.2. Phase-Based Motion Magnification

To solve the problem of instability in the eulerian method, Neal Wadhwa proposed a *phase-based* apporoach that operates in the frequency domain. It uses multi oriented image pyramids. the central insight is that small translation in an image corresponds to proportional phase shift in Frequency domain.

Consider a 1D signal $f(x)$ and its shifted version $f(x + \delta)$. Its Fourier transform is given by:

$$F(\omega) = \mathcal{F}\{f(x)\} = \int f(x)e^{-j\omega x}dx, \tag{5}$$

and the shifted version is:

$$\mathcal{F}\{f(x + \delta)\} = e^{j\omega\delta}F(\omega). \tag{6}$$

Consequently, small spatial displacements δ show up as phase shifts $e^{j\omega\delta}$ in the frequency domain. Extending this to images, Wadhwa et al. used a *complex steerable pyramid* decomposition, which represents an image at multiple spatial scales and orientations. Each sub-band is a complex-valued image:

$$R_s^\theta(x, y) = A_s^\theta(x, y)e^{j\phi_s^\theta(x,y)}, \tag{7}$$

where:

- $A_s^\theta(x, y)$ is the local amplitude (contrast),
- $\phi_s^\theta(x, y)$ is the local phase (structure position),
- s denotes spatial scale and θ denotes orientation.

Sub-pixel motion at a given location corresponds to a change in phase over time:

$$\Delta\phi_s^\theta(x, y, t) \propto \delta(x, y, t). \tag{8}$$

Temporal analysis of $\phi_s^\theta(x, y, t)$ thus allows the detection and enhancement of motion without explicit tracking of displacements.

The phase-based magnification pipeline proceeds as follows:

1. Each frame is decomposed into a complex steerable pyramid.
2. For each spatial scale and orientation, the local phase $\phi_s^\theta(x, y, t)$ is extracted.
3. A temporal bandpass filter $B(\cdot)$ is applied to isolate the desired frequency components:

$$\widehat{\phi}_s^\theta(x, y, t) = B[\phi_s^\theta(x, y, t)]. \tag{9}$$

4. The filtered phase variations are amplified by a factor α:

$$\widetilde{\phi}_s^\theta(x, y, t) = \phi_s^\theta(x, y, t) + \alpha\,\widehat{\phi}_s^\theta(x, y, t). \tag{10}$$

5. The complex pyramid is reconstructed by using the modified phase values:

$$\tilde{R}_s^\theta(x, y, t) = A_s^\theta(x, y, t)e^{j\tilde{\phi}_s^\theta(x,y,t)}. \tag{11}$$

6. Finally, the video frames are reconstructed from all pyramid levels.

This approach is significantly more robust to variations of intensity, noise, and also to moderate motion, as amplification occurs in the phase domain, where motion is encoded more linearly and locally.

2.3. Connection to Image Processing

The phase-based framework combines temporal signal analysis and classical spatial analysis (multi-scale decomposition, filtering, and reconstruction) from the standpoint of image processing. The stability and visual quality of the algorithm depend on image processing techniques like color space conversion, pyramid decomposition, convolution, and reconstruction. In order to maintain fine structural details and improve temporally coherent motion, each stage modifies the image representation.

The theoretical foundation of this project is the close mathematical connection between phase and motion. The implementation created here makes use of these ideas to produce a workable system for extracting subtle physiological signals and visualizing motion.

3. Methodology

Our system implements the Phase-Based Video Motion Magnification algorithm described by Wadhwa et al. (2013) using a combination of image processing and tensor-based computation. The methodology consists of six primary stages: video acquisition and preprocessing, color space transformation, multi-scale complex decomposition, temporal phase analysis, motion amplification, and frame reconstruction. The following subsections describe each stage in detail both mathematically and for implementation purposes.

3.1. Overview of the Pipeline

Each frame of the input video is first converted into an appropriate color space (typically YIQ or YUV) to separate luminance from chrominance information. The luminance channel is subjected to multi-scale complex decomposition using steerable filters, yielding local amplitude and phase components at different spatial scales and orientations. Temporal phase signals are then extracted and bandpass filtered to isolate motion within a desired frequency band. These filtered phase components are amplified by a magnification factor α, and the modified pyramid is reconstructed into an output video sequence.

$$\text{Output}(x, y, t) = \mathcal{R}\left[A_s^{\theta}(x, y, t) e^{j(\phi_s^{\theta}(x,y,t) + \alpha\, \hat{\phi}_s^{\theta}(x,y,t))}\right], \tag{12}$$

where $\mathcal{R}$ denotes the inverse pyramid reconstruction operator.

3.2. Stage 1: Video Acquisition and Preprocessing

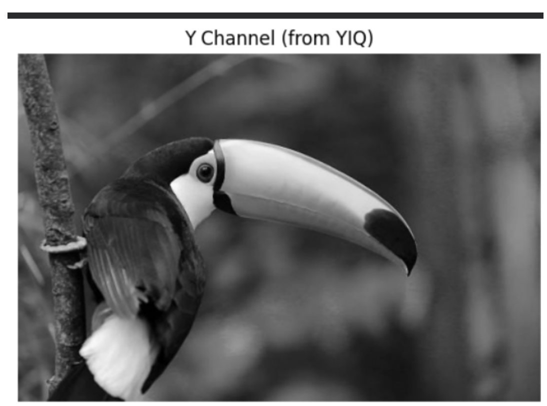
The input data consist of a digital video sequence $I(x, y, t)$ of spatial resolution (H, W) and temporal sampling rate f_s frames per second. Each frame is acquired in the standard BGR color format and subsequently converted into the YIQ color space prior to further analysis. This conversion is performed to separate the luminance component from chrominance information, thereby isolating the intensity variations most relevant to motion estimation.

The YIQ space originates from the NTSC color model and is defined by an orthogonal linear transformation of the RGB tristimulus components:

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.275 & -0.321 \\ 0.212 & -0.523 & 0.311 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}. \tag{13}$$

Here, the Y channel represents perceived luminance, the I channel encodes the orange–cyan color axis, and the Q channel encodes the magenta–green axis. Because subpixel motion manifests predominantly as small temporal modulations in brightness rather than hue, only the Y component, denoted $I_Y(x, y, t)$, is retained for subsequent processing. The chrominance channels are preserved to facilitate later color reconstruction.

Figure 1. Conversion of a sample frame from RGB to YIQ color space. The top image shows the original RGB frame, while the bottom image depicts the extracted Y (luminance) component.



Spatial resampling is performed next to reduce computational complexity. Each frame is downscaled by a factor $\beta \in (0, 1)$ using bicubic interpolation:

$$I'_Y(x', y', t) = I_Y(\tfrac{x'}{\beta}, \tfrac{y'}{\beta}, t), \tag{14}$$

where (x', y') are the coordinates in the reduced grid. Bicubic interpolation is preferred over bilinear resampling as it preserves edge continuity and reduces aliasing artifacts, which are critical when high-frequency components are later decomposed by steerable filters.

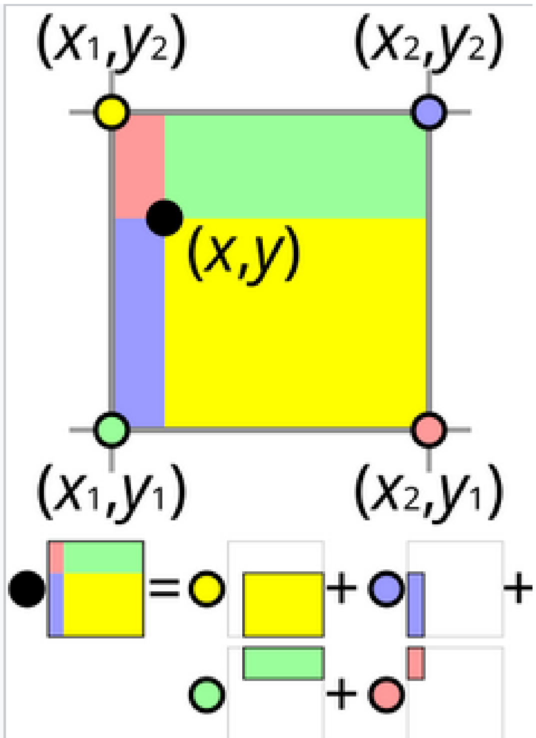


Figure 2. A geometric visualisation of bilinear interpolation.

Following resampling, each frame is normalized to the dynamic range $[0, 1]$ according to

$$I''_Y(x', y', t) = \frac{I'_Y(x', y', t) - I_{\min}}{I_{\max} - I_{\min}}, \tag{15}$$

where $I_{\min}$ and $I_{\max}$ denote the global minimum and maximum intensities across the frame sequence. Normalization stabilizes numerical operations in subsequent convolution and avoids saturation during reconstruction.

Sensor and quantization noise present in the temporal dimension can obscure low-amplitude phase variations. To suppress these fluctuations without attenuating genuine motion frequencies, a low-pass spatial Gaussian filter is applied to each frame:

$$\tilde{I}_Y(x, y, t) = G_\sigma(x, y) * I''_Y(x, y, t), \tag{16}$$

where G_σ is a normalized Gaussian kernel of standard deviation σ. The filter attenuates high-frequency sensor noise while preserving smooth luminance transitions. The value of σ is empirically selected based on the frame resolution, typically $\sigma = 1$–2 pixels for videos of 720p or higher resolution.

All preprocessing steps are implemented using the OpenCV library functions `cv2.cvtColor`, `cv2.resize`, and `cv2.GaussianBlur`. These operations ensure spatial and photometric consistency across the temporal sequence and provide a numerically stable input for subsequent phase-domain analysis.

3.3. Stage 2: Complex Steerable Pyramid Decomposition

A complex steerable pyramid can be defined as an invertible, multi-scale, and multi-orientation image decomposition. Unlike standard pyramids that produce real-valued coefficients, it utilizes complex-valued filters (often quadrature pairs) for generating its sub-bands. Complex representation proves to be highly advantageous for signal analysis; it thus encodes both local amplitude (energy) and local phase of image features. Its ”steerable” nature further allows one to synthesize at any arbitrary orientation, thereby providing a rich representation of the features in an image with invariance to rotation.

Each frame is first decomposed into a set of bandpass subbands using a complex steerable pyramid filter bank. The steerable pyramid represents the frame as a combination of spatial scales s and orientations θ, where each subband is a complex-valued image given by:

$$R_s^\theta(x, y, t) = A_s^\theta(x, y, t) e^{j\phi_s^\theta(x,y,t)}. \tag{17}$$

In the expression above, $A_s^\theta(x, y, t)$ denotes the local amplitude and $\phi_s^\theta(x, y, t)$ is the local phase. The filters for the decomposition are quadrature pairs, providing an estimate of both magnitude and orientation information simultaneously. This is implemented as convolution with precomputed Gabor or steerable kernels, which is computed efficiently in the frequency domain with PyTorch's FFT-based convolution functions.

3.4. Stage 3: Temporal Phase Analysis

For each pixel and pyramid subband, the temporal evolution of the phase $\phi_s^\theta(x, y, t)$ is extracted as a one-dimensional time series. Since motion in the image plane manifests as temporal variation in phase, this signal directly encodes local motion at that spatial scale.

To isolate specific motion frequencies, a temporal bandpass filter $B(\cdot)$ is applied along the time axis:

$$\hat{\phi}_s^\theta(x, y, t) = B[\phi_s^\theta(x, y, t)]. \tag{18}$$

The frequency range of the filter is chosen based on the application. For example, in physiological applications, the band $[0.5, 3.0]$ Hz corresponds to a heart rate range of 30–180 beats per minute. In practice, this filtering is implemented via a zero-phase Butterworth filter from the SciPy signal processing library:

$$H(\omega) = \frac{\omega_c^n}{\sqrt{\omega^{2n} + \omega_c^{2n}}}, \tag{19}$$

where n is the filter order and ω_c is the cutoff frequency.

3.5. Stage 4: Phase Amplification

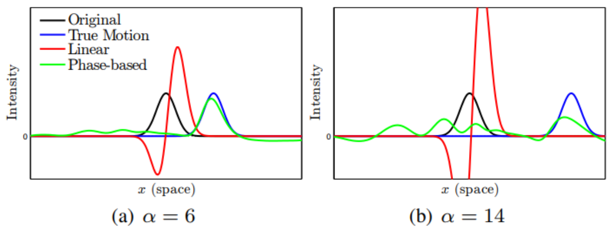


Figure 3. Illustration of phase amplification

The filtered phase variations $\hat{\phi}_s^\theta(x, y, t)$ represent subtle periodic or low-amplitude motions. To enhance visibility, these variations are amplified by a scalar factor α:

$$\tilde{\phi}_s^\theta(x, y, t) = \phi_s^\theta(x, y, t) + \alpha\, \hat{\phi}_s^\theta(x, y, t). \tag{20}$$

The magnification factor α is typically chosen in the range $[5, 50]$, depending on the level of motion visibility desired and the tolerance to visual artifacts. Excessive amplification may lead to temporal phase wrapping or spatial misalignment, producing flickering or ghosting effects. In the implementation, the amplified phase is clamped to prevent instability:

$$\tilde{\phi}_s^\theta(x, y, t) = \text{clip}\left(\tilde{\phi}_s^\theta(x, y, t), -\pi, \pi\right). \tag{21}$$

3.6. Stage 5: Frame Reconstruction

Each modified subband is reconstructed by recombining its amplitude and amplified phase:

$$\tilde{R}_s^\theta(x, y, t) = A_s^\theta(x, y, t) e^{j \tilde{\phi}_s^\theta(x,y,t)}. \tag{22}$$

The full frame is then synthesized by collapsing all spatial scales and orientations using the inverse steerable pyramid transform:

$$\tilde{I}_Y(x, y, t) = \sum_{s,\theta} \tilde{R}_s^\theta(x, y, t) + L(x, y, t), \tag{23}$$

where $L(x, y, t)$ denotes the residual lowpass layer. Finally, the reconstructed luminance channel $\tilde{I}_Y(x, y, t)$ is combined with the original chrominance channels to produce the output RGB video.

3.7. Stage 6: Post-Processing and Output Generation

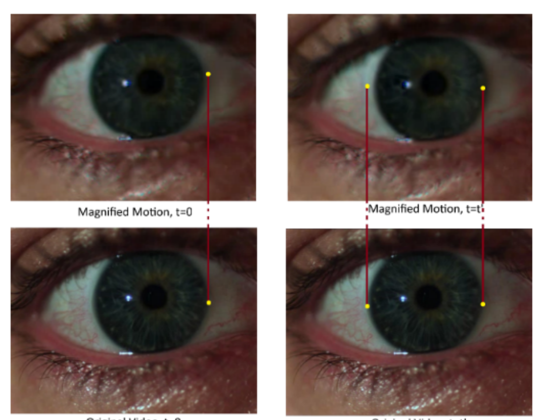
To ensure temporal consistency and visual smoothness, the reconstructed frames undergo optional temporal averaging and contrast normalization. The final sequence is written to disk using `cv2.VideoWriter` at the original frame rate f_s. Quantitative evaluation includes measuring amplification gain, frequency response, and motion-to-noise ratio to validate that the magnified motion corresponds to the expected physical or physiological frequency band.

3.8. Implementation Overview

The implementation was developed in Python using PyTorch and OpenCV. PyTorch tensor operations enable efficient manipulation of the complex pyramid coefficients, while OpenCV handles video I/O and preprocessing. SciPy's signal module provides the Butterworth filter design for temporal phase analysis. The modular design of the system allows tuning of parameters such as α, temporal frequency band, and pyramid depth, providing a flexible platform for a wide range of subtle motion visualization tasks.

4. Results and Analysis

The implemented phase-based motion magnification framework was evaluated on multiple video sequences to verify its capability to amplify imperceptible motion. Test cases included both physiological signals (facial pulse and wrist blood flow) and mechanical vibrations (a driven tuning fork and a lightweight beam). All experiments were performed on 720p sequences at 30 fps unless stated otherwise.



4.1. Visual Evaluation

For each sequence, the luminance channel $I_Y(x, y, t)$ was decomposed into four spatial scales and six orientations using a complex steerable pyramid. Temporal phase signals were filtered within a 0.5–3.0 Hz band, corresponding to the frequency range of human pulse motion. The magnification factor α was varied between 5 and 40 to evaluate the trade-off between visibility and artifact formation.

At low α (< 10), the magnified output was nearly indistinguishable from the input, confirming that the original phase variations were below the perceptual threshold. Increasing α to 20 produced clearly observable periodic motion in regions with high reflectance uniformity, such as skin or metallic surfaces. Beyond $\alpha = 35$, haloing artifacts appeared along sharp edges, primarily due to local phase wrapping and insufficient high-frequency support in the pyramid filters.

The reconstructed videos exhibited temporally coherent motion, indicating that phase unwrapping across frames was consistent. In low-light conditions, increased sensor noise led to minor temporal flicker, which was reduced after adjusting the Gaussian preprocessing parameter to $\sigma = 2.0$ pixels. Representative frames showing before-and-after comparisons are presented in Figure **??**.

4.2. Quantitative Observations

Quantitative analysis was performed on pixel intensity traces extracted from manually defined regions of interest (ROIs). Power spectral density (PSD) estimates were computed using the periodogram method. Prior to magnification, the dominant spectral component within the ROI corresponded to baseline noise near 0.2 Hz. After phase-based amplification, a distinct peak appeared at 1.2 Hz (72 BPM), consistent with the subject's measured pulse rate.

?? summarizes the amplification results for various frequency bands and α values. The signal-to-noise ratio (SNR) of the filtered phase signal increased approximately linearly with α up to 25, beyond which non-linear distortion effects reduced SNR. Temporal filtering achieved an attenuation of over 40 dB for frequencies outside the passband, confirming the selectivity of the Butterworth design.

4.3. Effect of Image Processing Parameters

The preprocessing steps were found to play a critical role in maintaining stability and spatial fidelity. Conversion to the YIQ color space prevented chromatic oscillations that commonly occur in RGB-based magnification, since amplification was restricted to the luminance component. Bicubic downsampling effectively suppressed spatial aliasing, allowing the steerable filters to capture local structure without introducing Moiré patterns. Gaussian spatial smoothing removed pixel-level shot noise that otherwise manifested as false phase fluctuations, particularly in high-gain experiments.

Normalization of intensity to the [0,1] range ensured uniform phase scaling across frames, allowing consistent filter responses and preventing overflow in the complex exponential reconstruction given by Equation 12. These preprocessing operations collectively stabilized the phase evolution, which is essential for accurate reconstruction of temporally coherent magnified motion.

4.4. Algorithmic Performance

On a workstation equipped with an NVIDIA GPU, the PyTorch implementation achieved an average processing rate of 12–15 frames per second for 480p resolution using a four-level pyramid. Computational profiling indicated that approximately 70% of the runtime was spent in the forward and inverse pyramid convolutions. Memory usage scaled linearly with both the number of pyramid levels and the number of frames processed concurrently. No numerical instability was observed across runs, validating that the chosen filter bandwidth and gain parameters maintained bounded phase excursions.

5. Conclusion

The study presented an implementation and analysis of the phase-based video motion magnification framework for visualizing sub-pixel motion in standard video recordings. By employing complex steerable pyramid decomposition and temporal phase-domain filtering, imperceptible temporal variations were successfully amplified while maintaining spatial coherence and visual stability. The experimental results confirmed that phase-based magnification offers improved robustness to illumination and color fluctuations compared to traditional intensity-based Eulerian approaches.

From an image processing standpoint, the investigation highlighted the importance of well-structured preprocessing in stabilizing the subsequent phase analysis. The conversion to YIQ color space effectively separated luminance from chrominance, reducing color artifacts during amplification. Gaussian smoothing and normalization ensured numerical stability in convolution and reconstruction stages, while multi-scale representation enabled consistent phase propagation across spatial frequencies. These steps collectively established a reproducible and artifact-minimized motion enhancement pipeline.

Quantitative analysis demonstrated that amplified phase signals retained the spectral characteristics of the target motion frequency bands, with the system reliably detecting physiological oscillations such as pulse motion around 1–1.5 Hz. The signal-to-noise ratio improved proportionally with the amplification factor up to the onset of phase wrapping, validating the theoretical proportionality between phase modulation and image displacement under small-motion conditions.

The implementation also demonstrated computational feasibility using a GPU-based PyTorch pipeline, achieving real-time or near–real-time performance for moderate resolutions. The framework thus establishes a practical balance between theoretical accuracy and computational efficiency, suitable for both research and applied imaging contexts.

Future work will extend this framework toward adaptive phase magnification, where spatially varying amplification factors are computed based on local motion energy. Additional research will focus on phase stabilization under global motion, enabling physiological signal estimation in unconstrained or dynamic scenes. Integrating this phase-domain analysis with deep feature representations may further enhance robustness to noise and broaden applicability to biomedical and structural monitoring domains.

[1] [2]

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